

COMPUTER NETWORKS – HANDWRITTEN EXAM NOTES

UNIT 1: Introduction to Networks, LAN/MAN/WAN, Topologies, OSI & TCP/IP Models, Transmission Media

B.Tech Mid-1 Examination Preparation | Complete Question Bank Solutions

PART A – 2 MARKS QUESTIONS

Question 1

Define a computer network

Definition

A **Computer Network** is an interconnected collection of autonomous computing devices (computers, servers, switches, routers, IoT devices) that communicate with each other over transmission media using a standardized set of communication rules called **protocols** to share data, hardware (printers, storage), and software resources.

Key Characteristics

- **Resource Sharing:** Centralized sharing of files, databases, and expensive peripherals.
- **Autonomy:** No single computer exerts absolute master-slave control over another independent host.
- **Reliability:** Alternative communication paths ensure data delivery even if a link fails.

Question 2

What are LAN, MAN and WAN?

1. **LAN (Local Area Network):** Operates within a small, limited geographical area (up to 1–2 km) such as a single room, office, building, or college campus. Owned privately with high data transfer rates (100 Mbps to 10 Gbps) and very low error rates. (e.g., College Computer Lab Ethernet/Wi-Fi).
2. **MAN (Metropolitan Area Network):** Spans a larger geographical area than a LAN, typically covering an entire town or city (5 to 50 km). Often owned by a consortium or single network provider. (e.g., City Cable TV network, Municipal Wi-Fi).
3. **WAN (Wide Area Network):** Covers broad regional, national, or global distances (> 100 km), connecting multiple LANs and MANs across countries and continents using public telecommunication links, undersea fiber cables, and satellites. (e.g., The Internet).

Question 3

List the different types of network topologies

Network Topology defines the geometric and physical/logical arrangement of communication nodes and links in a network:

1. **Bus Topology:** All nodes connect to a single central transmission line (backbone bus) terminated at both ends.
2. **Star Topology:** Every node connects individually to a central controller (Hub/Switch) via point-to-point links (most widely used in modern LANs).
3. **Ring Topology:** Each device connects to exactly two adjacent neighbors, forming an unbroken circular unidirectional loop.
4. **Mesh Topology:** Every device has a dedicated point-to-point physical connection to every other device in the network (Number of links = $n(n-1)/2$).
5. **Tree Topology:** Hierarchical combination of star and bus topologies where groups of star-configured networks are connected to a central bus backbone.
6. **Hybrid Topology:** Combination of two or more different topologies (e.g., Star-Ring, Star-Bus).

Question 4

Explain the purpose of the OSI Reference Model

Purpose and Significance

The Open Systems Interconnection (OSI) reference model was established by the International Organization for Standardization (ISO) in 1984. Its primary purposes are:

- **Open Vendor Neutrality:** Enables heterogeneous hardware and operating systems from different vendors to communicate seamlessly.
- **Layered Modularity:** Decomposes complex network communication into 7 distinct, manageable functional layers with well-defined service interfaces.
- **Independent Evolution:** Modifications or upgrades within one layer (e.g., switching from IPv4 to IPv6) do not require changes in other layers.
- **Standardized Education & Troubleshooting:** Provides an industry-standard conceptual foundation for designing, teaching, and diagnosing network protocols.

Question 5

What is the difference between guided and unguided transmission media?

Feature	Guided Media (Bounded / Wired)	Unguided Media (Unbounded / Wireless)
Physical Path	Signals are directed and confined within a solid physical conductor/cable.	Signals propagate freely through air, vacuum, or water using electromagnetic waves.
Security	High security; tapping requires physical access to the wire.	Lower security; broadcast signals can be intercepted/eavesdropped easily.
Interference	Protected by shielding; lower electromagnetic interference.	Susceptible to atmospheric noise, obstacles, and multipath fading.
Mobility	Fixed installation; no physical user mobility during transmission.	High flexibility; supports mobile users seamlessly.
Examples	Twisted-pair cable, Coaxial cable, Optical Fiber cable.	Radio waves, Microwaves, Infrared, Satellite communication.

----- PART B – 5 MARKS / ESSAY QUESTIONS -----

Essay Question 1

Explain the different types of computer networks: LAN, MAN and WAN with suitable examples

1. Introduction

Computer networks are classified primarily on the basis of their **geographical scope**, data transmission rates, ownership, and structural architecture into three major categories: **LAN**, **MAN**, and **WAN**.

2. Detailed Explanation of Network Types

(a) LAN (Local Area Network)

- **Geographical Coverage:** Confined to a single room, building, laboratory, or campus (typically $\leq 1-2$ km radius).
- **Ownership & Control:** Owned and managed privately by a single individual, institution, or organization.
- **Data Rate:** Very high transmission speed, ranging from 100 Mbps (Fast Ethernet) to 1 Gbps / 10 Gbps (Gigabit Ethernet).
- **Error Rate & Delay:** Extremely low bit-error rate ($\sim 10^{-9}$) and minimal propagation delay due to short physical distances.
- **Technologies & Topologies:** IEEE 802.3 Ethernet (Twisted Pair/Fiber), IEEE 802.11 Wi-Fi; Star, Bus, or Ring topologies.

- **Example:** A college computer lab network, an office Wi-Fi network, a home broadband network connecting smartphones and laptops.

(b) MAN (Metropolitan Area Network)

- **Geographical Coverage:** Encompasses an entire city or large municipal region (5 km to 50 km).
- **Ownership:** Often owned by a consortium of users, a telecom service provider, or a municipal government.
- **Data Rate:** Moderate to high data rates (typically 100 Mbps to 1 Gbps) utilizing fiber optic ring backbones.
- **Technologies:** DQDB (Distributed Queue Dual Bus – IEEE 802.6), Metro Ethernet, SONET/SDH, WiMAX (IEEE 802.16).
- **Example:** City Cable TV distribution networks, interconnected branches of a bank across a city, municipal Smart City surveillance network.

(c) WAN (Wide Area Network)

- **Geographical Coverage:** Spans vast geographical territories across states, nations, continents, or the entire planet (> 100 km).
- **Ownership:** Publicly or collectively owned; relies on infrastructure leased from telecommunication common carriers.
- **Transmission Media:** Fiber-optic undersea cables, high-power microwave relay towers, and geostationary/LEO satellite links.
- **Error Rate & Delay:** Higher error rates and significant propagation delay due to vast physical distances and multiple routing hops.
- **Technologies:** IP routing, MPLS, Frame Relay, ATM, BGP, Satellite links.
- **Example:** The Global Internet, multinational enterprise networks (e.g., Google or Microsoft private global backbones).

3. Comprehensive Comparison Table

Parameter	LAN	MAN	WAN
Geographic Span	Small (up to 1–2 km)	Moderate (5–50 km, city-wide)	Vast (> 100 km, global)
Ownership	Private	Private or Public Provider	Distributed / Telecom Provider
Speed / Bandwidth	Highest (100 Mbps – 10 Gbps)	Moderate (44 Mbps – 1 Gbps)	Lower/Variable (10 Mbps – 100 Gbps)
Propagation Delay	Shortest (microseconds)	Moderate (milliseconds)	Longest (tens/hundreds of ms)
Error Rates	Lowest	Moderate	Highest

Cost of Setup	Low	High	Very High
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Conclusion: LAN provides high-speed private local communication, MAN interconnects multiple LANs across a municipality, and WAN provides the global fabric connecting all networks worldwide.

Essay Question 2

Describe the OSI Reference Model and explain the functions of its layers

1. Introduction to the OSI Model

The OSI (Open Systems Interconnection) Model is a conceptual 7-layer architectural framework standardized by the ISO (International Organization for Standardization) in 1984 under standard ISO 7498. It provides a universal blueprint defining how data moves from a software application on one computer across a network medium to an application on another computer.

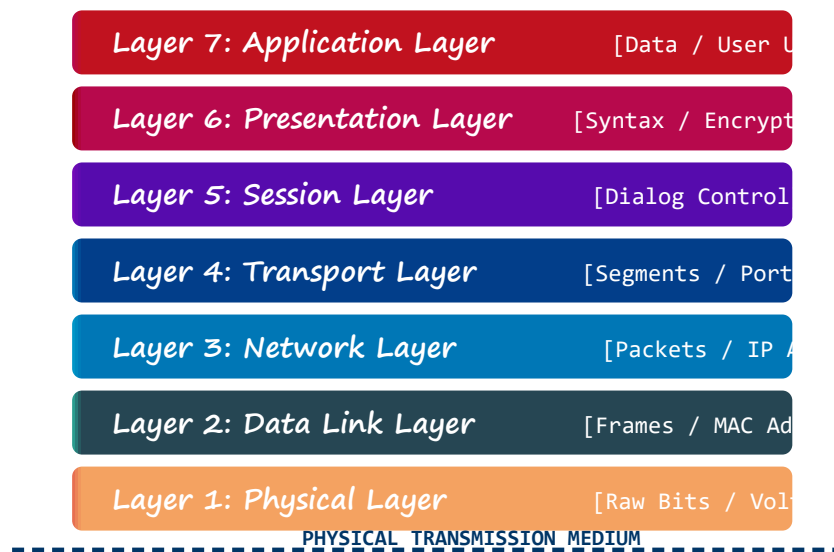


Figure 1.1: The 7 Layers of the ISO-OSI Reference Model

2. Detailed Functions of the 7 Layers (Top to Bottom)

1. Physical Layer (Layer 1)

- **Data Unit:** Raw Bits (0s and 1s).
- **Functions:** Defines electrical signals, voltages, transmission rates (bits per second), physical topology, and mechanical connectors (e.g., RJ-45).
- **Transmission Modes:** Simplex, Half-Duplex, and Full-Duplex.
- **Protocols & Hardware:** Hubs, Repeaters, RS-232, Cat-6 cables.

2. Data Link Layer (DLL) (Layer 2)

- **Data Unit:** Frames.
- **Functions:**
 1. **Framing:** Encapsulates packets into distinct frames with headers and trailers.
 2. **Physical Addressing:** Adds 48-bit physical MAC addresses of source and destination.
 3. **Flow Control:** Prevents a fast sender from overwhelming a slow receiver.
 4. **Error Control:** Detects and retransmits damaged/lost frames using CRC and ARQ.
 5. **Media Access Control (MAC):** Governs which device transmits on a shared channel (CSMA/CD).
- **Hardware & Protocols:** Switches, Bridges, Ethernet (802.3), Wi-Fi (802.11), PPP.

3. Network Layer (Layer 3)

- **Data Unit:** Packets.
- **Functions:**
 1. **Logical Addressing:** Assigns global IP addresses (IPv4 / IPv6) to uniquely identify source and destination across networks.
 2. **Routing:** Selects the best path for data using routing algorithms (OSPF, BGP, RIP).
 3. **Packet Switching & Fragmentation:** Breaks packets down to fit MTU limits.
- **Hardware & Protocols:** Routers, Layer 3 Switches, IP, ICMP, ARP, IGMP.

4. Transport Layer (Layer 4)

- **Data Unit:** Segments.
- **Functions:**
 1. **End-to-End (Process-to-Process) Delivery:** Delivers data directly to specific applications using Port Numbers (e.g., Port 80 for HTTP).
 2. **Segmentation & Reassembly:** Divides application data into segments and reorders them at destination.
 3. **Connection Management:** Provides reliable connection-oriented service (TCP) or lightweight connectionless service (UDP).
 4. **Congestion Control:** Regulates transmission rate to prevent network bottlenecks.
- **Protocols:** TCP, UDP, SCTP.

5. Session Layer (Layer 5)

- **Functions:** Establishes, maintains, synchronizes, and terminates communication sessions between applications.
- **Dialog Control:** Manages who talks and when (simplex, half-duplex, or full-duplex).
- **Synchronization / Checkpointing:** Inserts checkpoints into data streams so that in case of failure, retransmission resumes from the last checkpoint rather than the beginning.
- **Protocols:** NetBIOS, RPC, PPTP.

6. Presentation Layer (Layer 6)

- **Functions:** Manages syntax and semantics of the transmitted information:
 1. **Translation / Encoding:** Translates between different encoding standards (e.g., ASCII to EBCDIC).
 2. **Encryption & Decryption:** Secures sensitive data (e.g., SSL/TLS encryption).
 3. **Compression:** Reduces the number of bits to be transmitted (JPEG, MPEG, ZIP).

7. Application Layer (Layer 7)

- **Functions:** Directly interfaces with human users and software applications. Provides network services like Web browsing, Email transfer, File access, and Directory lookups.
- **Protocols:** HTTP, HTTPS, FTP, SMTP, DNS, DHCP, Telnet, SSH.

3. Data Encapsulation & Decapsulation Principle

As data moves *down* the stack from Application to Physical, each layer appends its own control header (and trailer at Layer 2) — a process called **Encapsulation**. At the receiver, as signals move *up* the stack, each layer strips its corresponding header — a process called **Decapsulation**.

Essay Question 3

Explain the TCP/IP Reference Model and compare it with the OSI Reference Model

1. Introduction to TCP/IP Model

The TCP/IP (Transmission Control Protocol / Internet Protocol) Model, originally developed by the DoD (Department of Defense, USA) in the 1970s for ARPANET, is the practical, working architectural model of the modern Internet. It groups networking protocols into **4** functional layers:

2. The 4 Layers of the TCP/IP Model

1. **Application Layer:** Combines the functions of OSI Application, Presentation, and Session layers. Handles high-level protocols, data formatting, encryption, and user interfaces. (Protocols: HTTP, HTTPS, FTP, SMTP, DNS, SSH, DHCP).
2. **Transport Layer (Host-to-Host):** Responsible for end-to-end process-to-process communication. Provides two primary protocols:
 - **TCP (Transmission Control Protocol):** Reliable, connection-oriented, ordered byte-stream delivery with flow and congestion control.
 - **UDP (User Datagram Protocol):** Unreliable, connectionless, low-overhead best-effort datagram delivery.

3. **Internet Layer:** Corresponds to OSI Network Layer. Handles packet routing, logical addressing, and fragmentation across multiple independent networks. (Protocols: IP (IPv4/IPv6), ICMP (diagnostics/ping), ARP (IP to MAC mapping), IGMP).
4. **Network Access Layer (Host-to-Network):** Combines OSI Data Link and Physical layers. Interfaces directly with physical transmission hardware and manages framing, MAC addressing, and signal transmission over copper, fiber, or wireless media. (Protocols: Ethernet 802.3, Wi-Fi 802.11, PPP).

3. Detailed Comparison: OSI vs TCP/IP

Aspect	OSI Reference Model	TCP/IP Reference Model
Number of Layers	7 Layers (Physical, DLL, Network, Transport, Session, Presentation, Application)	4 Layers (Network Access, Internet, Transport, Application)
Nature / Purpose	Theoretical reference model (standardized before protocols were built)	Practical implementation model (protocols built first, model described them)
Service / Protocol Separation	Strictly distinguishes between Services, Interfaces, and Protocols	Does not cleanly separate services and protocols
Network Layer Approach	Supports both Connectionless and Connection-oriented communication	Supports only Connectionless communication (IP is strictly connectionless)
Transport Layer Approach	Supports only Connection-oriented communication	Supports both Connection-oriented (TCP) and Connectionless (UDP)
Session & Presentation	Distinct, dedicated layers	Merged entirely into the Application Layer
Usage & Dominance	Used mainly as a teaching guide and conceptual standard	The actual protocol suite powering the global commercial Internet
Protocol Replacement	Protocols are easily replaced due to strict layer independence	Protocols are not easily replaced (hard-tied to IP and TCP architectures)

Essay Question 4

Explain guided transmission media with reference to twisted-pair cable, coaxial cable and fiber-optic cable

1. Introduction to Guided Media

Guided (Bounded / Wired) Transmission Media physically channel and confine electromagnetic waves within a tangible physical conductor across the network path. The

three primary types used in computer networks are **Twisted-Pair Cable**, **Coaxial Cable**, and **Fiber-Optic Cable**.

2. Detailed Technical Study of the Three Media

(a) Twisted-Pair Cable

- **Physical Structure:** Consists of two insulated copper wires (typically 1 mm thick) twisted together in a regular helical pattern.
- **Why Twisting?** Twisting ensures both wires are equally exposed to external electromagnetic interference (EMI) and crosstalk. In differential signaling, the receiver subtracts the voltages on both wires, canceling out the induced noise!
- **Types:**
 1. **UTP (Unshielded Twisted Pair):** Common in indoor Ethernet LANs (Cat 5e, Cat 6, Cat 6a). Low cost, highly flexible, uses **RJ-45** connectors. Max segment length: 100 meters.
 2. **STP (Shielded Twisted Pair):** Each pair is wrapped in metal foil with an overall braided wire shield. Provides superior noise immunity in noisy industrial plants, but is bulkier and more expensive.
- **Data Rate & Applications:** 10 Mbps (Cat 3) up to 10 Gbps (Cat 6a). Widely used in telephone lines and desktop Ethernet connections.

(b) Coaxial Cable ("Coax")

- **Physical Structure:** Consists of 4 concentric layers:
 1. Central solid/stranded Copper Conductor (carries the signal).
 2. Dielectric Plastic Insulator.
 3. Outer Braided Copper/Aluminum Conductor Shield (serves as ground and shields against external EMI).
 4. Outer Protective Plastic Jacket.
- **Types:**
 - **50-ohm Coax:** Used for digital transmission (e.g., 10Base5 "Thicknet", 10Base2 "Thinnet" with BNC connectors).
 - **75-ohm Coax (RG-6, RG-59):** Used for analog transmission and broadband cable television / Cable Internet modems.
- **Bandwidth & Attenuation:** Higher bandwidth and better noise immunity than twisted pair over longer distances, but stiffer and more expensive to install.

(c) Fiber-Optic Cable

- **Physical Structure:** Transmits information as pulses of light (photons) through high-purity glass/silica fibers:
 1. **Core:** Ultra-pure cylindrical glass/plastic core with refractive index n_1 .

2. **Cladding:** Surrounding glass layer with slightly lower refractive index n_2 ($n_1 > n_2$).

3. **Buffer Coating & Outer Jacket:** Kevlar reinforcement and plastic jacket protecting against moisture and physical strain.

- **Working Principle (Total Internal Reflection – TIR):** When light enters the core at an angle greater than the critical angle ($\theta > \theta_c$), it cannot cross into the lower-index cladding and is completely reflected back into the core, propagating with minimal attenuation.

- **Propagation Modes:**

- **Single-Mode Fiber (SMF):** Extremely narrow core (8–10 μm). Light travels along a single direct path with zero modal dispersion. Used with laser diodes for long-distance telecom and undersea backbones (> 50–100 km).
- **Multi-Mode Fiber (MMF):** Wider core (50–62.5 μm). Multiple light rays bounce at different angles. Uses LEDs; limited to campus backbones (≤ 2 km) due to modal dispersion.

- **Key Advantages:**

- Massive bandwidth (> 100 Gbps to Tbps).
- Completely immune to Electromagnetic Interference (EMI) and radio noise.
- Extremely low attenuation (signals travel tens of kilometers without repeaters).
- Highest security (cannot be tapped without breaking the fiber and cutting the signal).

3. Comparative Analysis of Guided Media

Characteristic	Twisted Pair Cable	Coaxial Cable	Fiber-Optic Cable
Signal Carrier	Electrical voltage	Electrical voltage	Light pulses (photons)
Data Rate	10 Mbps – 10 Gbps	10 Mbps – 100 Mbps	10 Gbps – 100+ Gbps
Max Distance	100 m	500 m (Thicknet)	Tens to Hundreds of km
EMI Immunity	Moderate (poor for UTP)	Good	Complete / Immune
Cost & Installation	Cheapest, very easy	Moderate cost & flexibility	Expensive, requires skilled termination

----- LAST-MINUTE EXAM REVISION (UNIT 1) -----

Quick Memory Pointers & Layer Protocol Reference

OSI Layer	Primary Data Unit	Key Functions	Associated Protocols / Hardware
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7. Application	Data	User UI, network services	HTTP, HTTPS, FTP, SMTP, DNS, SSH
6. Presentation	Data	Translation, Encryption (SSL/TLS), Compression	ASCII, EBCDIC, JPEG, MPEG, TLS
5. Session	Data	Session control, Checkpointing, Dialog	NetBIOS, RPC, PPTP
4. Transport	Segment	Process-to-process delivery, Port addressing	TCP, UDP
3. Network	Packet	Logical IP addressing, Routing, Path selection	IP (IPv4/IPv6), ICMP, ARP, Routers
2. Data Link	Frame	Framing, MAC addressing, Error & Flow control	Ethernet (802.3), Wi-Fi (802.11), Switches
1. Physical	Bit	Bit transmission, electrical signaling, cable specs	Cables, Hubs, Repeaters, RJ-45